

ALFIE KUNZ

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KEY SKILLS: Programming: C#, Python, VB.NET, HLSL. Tools & Frameworks: Unity Engine, WinForms, Git, LaTeX, Tikz, Visual Studio.

Core Competencies: High Performance Computing, (GPU) Agent-based Modelling Mathematical Modelling, Machine Learning (NNs), Fluid Dynamics, Technical Writing.

EDUCATION

UNIVERSITY OF WARWICK: Full-Time Student of Mathematics & Physics (MMathPhys)

2023 – 2027

- ❖ Achieved an overall weighted average of 82.4% across Years 1-3:
- ❖ Year 1 Grade of 88.0% ('Top 5' in the Cohort). Year 2 Grade of 85.4% (placement pending). Year 3 Grade of 78.5% (placement pending).
- ❖ Years 1-2 Module breadth: Mathematical Analysis, Applied Modelling and Algebra, Mathematics of Asymptotics, Physics of Quantum Mechanics, Applied Mathematics, Fluids and Electrodynamics, Computational Modelling, Programming, and Algorithms.
- ❖ 3rd Year Specialism through modules such as Computational Modelling, Mathematical Modelling of PDEs, Fluid Dynamics, Neural Computing.

BECKFOOT SCHOOL: A-Level & GCSE Student

2016 – 2023

- ❖ A-Level Grades: A*A*A*A for Mathematics, Further Mathematics, Physics, Computer Science, EPQ (Astrophysics-based: "How Does a Neutron Star Remain a Stable Entity?"). #1 in Cohort. GCSE Grades: Ten Grade 9s, as well as a Distinction in Music BTEC.
- ❖ Highlights & Achievements: Founder & Treasurer of the School Chess Club, featured Alumni for Open Days & Showcases.

PROFILE:

Proactive computational researcher, ranked in the 'Top 5' of my university cohort, specialising in fluid dynamics, numerical simulations, and emergent behaviours in biophysics. Proven success in translating cutting-edge mathematical theory into commercial-grade, high-performance software. Inventive science communicator, puzzle designer, and believer in the idea that the most complex behaviours stem from elegantly simple rules.

PROFESSIONAL ATTRIBUTES:

Charismatic Leadership & Operational Initiative, Technical & Interpersonal Communication, Public Engagement, Mentorship & Coaching, Adaptability & High-Pressure Composure, Self-Direction, Creative Reasoning & Problem Solving.

PROJECTS

Drop Collisions, and their Collective Behaviour (in Collaboration with the **Met Office**)

2026 – 2027

University of Warwick (Associated with Prof James Sprittles, under the PX402 Physics Project Module)

- ❖ Currently undertaking a research project on computationally predicting the boundaries between bouncing and merging of liquid drops, by developing a reduced-order 'boundary integral' numerical model. These multiphase flow dynamics become crucial for studying cloud formation, and thus weather and climate modelling. Project In collaboration with partners including the **Met Office**, **NCAR**, and **Syngenta**.

APIC Computational Fluid Dynamics (CFD) Research Project & Simulation

2025 – 2026

University of Warwick (Associated with Prof Radu Cimpanu, under the MA369 Essay Module) (C#, HLSL, Python, Unity)

- ❖ Identified a novel Reynolds Number value of 5420 for the APIC CDF regime, in a Taylor-Green Vortex – justifies the report as a worthwhile contribution to the industry.
- ❖ Self-directed original CFD research, applying cutting-edge technologies in novel scenarios to investigate high-performance, momentum-conserving vortices.
- ❖ Independent derivation and implementation of a real-time Eulerian-Lagrangian APIC system (C#, Unity), using techniques such as Conjugate Gradient Method with MICT-0 Preconditioning, and GPU parallelism. Optimised rendering via Instanced Procedural Drawing (HLSL).
- ❖ Automated post-processing, data analysis, and visualisation pipelines in Python.
- ❖ Authored a rigorous, well-referenced technical essay on my contributions to the field, using LaTeX and Tikz figures. Demonstrated strong scientific communication by presenting my findings to two specialist Warwick professors (with no prior exposure to the implementation).

Chess Artificial Intelligence & Interface

2021 – Present

Beckfoot School A-Level Project (+ Self-Motivated Continuation)

(VB.NET, WinForms)

- ❖ Architected an 'International Master' Elo (~2850, as of v10.0), commercial-quality Chess AI project, designed a stylish chess-playing interface to support the AI (for example, through a hand-crafted opening book from millions of games) and user interactions.
- ❖ Robustly engineered and optimised a custom NegaMax heuristic AI algorithm, using techniques such as Alpha-Beta Pruning, Zobrist Hashing, Iterative Deepening, and Dynamic Aspiration Windows.
- ❖ Evaluated against a vast breadth of previous variations & models, by integrating models into a sophisticated, custom-made swiss-style benchmarking system with a 'plug & play' ecosystem. Includes a custom Zobrist hashing engine for robust game-ending detection, and colourful live visuals.
- ❖ Contributed an open-source, easy-to-use, and highly-customisable opening book builder (made from millions of open-source games) as a GitHub repository to support the engine. Contains features such as automatic PGN parsing, built-in move validation, and automatic negligible move pruning.
- ❖ Awarded 100% on the initial A-Level CS NEA; recognised by the project supervisor as the 'best he had seen' in his years of teaching.

Machine Learning & Biophysical Modelling Experiments

2023, 2024

Independent Research Projects

(C#, HLSL, VB.NET)

- ❖ **Emergence from Evolving Slime Trails:** Simulated millions of atomic agents in real-time on the GPU (massively parallelised as a compute shader) to conduct biophysical research on agent-based physarum transport networks under a 'Patlak-Keller-Segel'-type couple system, and their evolution into emergent ecosystems. Self-taught C# and HLSL, under the Unity Game Engine.
- ❖ **Hand-drawn Doodle & Digit Recognising Neural Network (NN):** Achieves ~98% accuracy on the MNIST dataset, and ~92% on the Google 'Quick Draw' database (12 categories) in under 2 minutes of training. Employs high-level ML concepts such as back propagation, regularisation, and overfitting regulation. Augments database images for strong resilience against user-drawn images.

Photography and Academic Web-based Portfolio

2025 - Present

Independent Research Project

(HTML, CSS, JS, Python)

- ❖ Overhauled and re-engineered a baseline template into a bespoke academic and photography portfolio, which serves as a home for both photography advertisement and career building.
- ❖ Developed custom JavaScript modules for dynamic JSON gallery rendering and hosting, AES-256 encrypted client-side delivery, custom non-linear phase modulation background, and integration with GitHub repositories.

OTHER AWARDS: 4-time Distinction & Gold Award Achiever in 'Bebras' Computing Challenge (top 10% nationally); 2-time Distinction holder in 'TCS Oxford Computing Challenge'. Two-time Winner of 'Beckfoot Young Musician of the Year' and performer at weddings; 16 years of self-taught piano.

EXPERIENCE & VOLUNTARY WORK

Teaching Assistant & GCSE Tutor: Beckfoot School

2021 - 2022, 2025

- ❖ Mentored lower ability Year 11 students in mathematics & science, involving one-on-one interpersonal support with strong client communication, and the development of personalised revision plans, to tackle 'maths anxiety' and providing inspirational 'lightbulb' moments.
- ❖ Shadowed staff, and worked with diverse students, to discern the strengths of approachability and patience in tackling knowledge gaps.
- ❖ Designed and independently delivered an A-level maths lesson on series convergence (extending the spec to introduce Ramanujan summation and Zeno paradoxes) - achieving strong student engagement, and reinforcing my ideals of inspiration via creative teaching.
- ❖ Reflected heavily on my own performance by proactively seeking and implementing feedback from experienced staff.

Physics Student Ambassador: University of Warwick

2024 - Present

- ❖ Led and assisted Live Chat events, Q&A panels, and Department & Campus Tours, for prospective students. Rapport-building and accommodation of different perspectives (developed through training workshops) to provide an exciting, interactive insight into university life.

Retail Sales Associate: Whittakers Schoolwear, Shipley

2023 - 2024

- ❖ Rapidly adapted to a high-intensity retail environment during peak summer season, whilst maintaining professionalism and composure. Expanded responsibilities to encompass till operations, database operations, online order management, and onboarding of new employees.

Chess Club Founder, Leader, Treasurer: Beckfoot School

2021 - 2023

- ❖ Founded and effectively scaled a school chess club to a weekly crowd of 70+ young people. Showed strong initiative and handling school resources. Involved the hiring of volunteers, organising class teaching and individual mentorship.
- ❖ Established a tournament structure to enable students to represent their school in regional tournaments, demonstrating a concrete legacy.